Impact of Screen time and Sleep Quality on Digital Eye strain and Hair loss among College Students

AIM:

To investigate the association between screen time, sleep quality, digital eye strain and hair loss among college students

Questionnaire

Section A: Demographic Details

a. Age
b. Gender
c. Department
d. Year of Study

Section B: Screen Time

a. On average, how many hours do you spend using digital devices (phone, laptop, tablet, etc.) each day?
b. Which digital device do you use the most?
c. Do you use digital devices within 30 minutes before going to bed?
d. Do you take regular breaks while using digital devices?

Section C: Sleep Quality

a. On average, how many hours do you sleep each night?
b. What time do you usually go to bed?
c. How would you rate your overall sleep quality?
d. How often do you wake up feeling refreshed?

Section D: Digital Eye Strain

a. How often do you experience eye strain after prolonged screen use?
b. Which of the following symptoms do you experience after prolonged screen use?
c. How often do you experience headaches after prolonged screen use?
d. Do your eyes feel tired during or after online classes or studying?

Section E: Hair Loss

a. Have you noticed increased hair fall in the past 6 months?
b. How would you describe your current hair loss?
c. Do you have a family history of hair loss?
d. How often do you feel stressed?

Section F: Lifestyle

a. How many days do you exercise in a week?
a. How often do you eat fruits and vegetables?
b. How much water do you drink per day?

Section G: Opinion

a. Do you think excessive screen time affects your sleep quality?
b. Do you think poor sleep contributes to increased hair loss?
c. Do you think excessive screen time contributes to digital eye strain?

Tasks

- Import dataset (python)
- Check missing values (python)
- Clean data (python)
- Frequency tables
- Statistical Methods
- Correlation (Screen Time, Sleep Hours, Hair Loss)
- Chi-square test
- Multiple Linear Regression
- Interpretation
- Conclusion

Import dataset, checking missing values (python)

```python
import pandas as pd

df = pd.read_csv(r"C:\Users\acer\OneDrive\Desktop\Dataset.csv")

print(df.head())

print(df.isnull().sum())

# Import pandas
import pandas as pd

# Load the dataset
df = pd.read_csv(r"C:\Users\acer\OneDrive\Desktop\Dataset.csv")

# Display the first 5 rows
print("FIRST 5 ROWS OF THE DATASET")
print(df.head())

# Display dataset information
print("\nDATASET INFORMATION")
print(df.info())

# Display the shape of the dataset
print("\nSHAPE OF THE DATASET")
print(df.shape)

# Check missing values
print("\nMISSING VALUES")
print(df.isnull().sum())

# Check duplicate rows
print("\nDUPLICATE ROWS")
print(df.duplicated().sum())
```

Cleaning data

```python
# Clean the data
# Replace missing values in the Symptoms column with "None"
df["Symptoms"] = df["Symptoms"].fillna("None")

# Remove duplicate rows (if any)
df = df.drop_duplicates()

# Check missing values after cleaning
print("\nMISSING VALUES AFTER CLEANING")
print(df.isnull().sum())

# Save the cleaned dataset
df.to_csv(r"C:\Users\acer\OneDrive\Desktop\Cleaned_Dataset.csv", index=False)

print("\nData cleaned successfully!")
```

The dataset was imported into Python using the Pandas library. Missing values and duplicate records were checked, and the missing values in the Symptoms column were replaced with "None" to clean the data. The cleaned dataset was then saved for further statistical analysis in R.

OUTPUT:

```
In [2]: %runfile C:/Users/acer/.spyder-py3/temp.py --wdir
   Age  Gender                   College  ... OpinionSleep OpinionHair OpinionEye
0   19    Male   Madras Christian College  ...          No          No      Maybe
1   18  Female  Women's Christian College  ...       Maybe          No      Maybe
2   23    Male     SRM IST Kattankulathur  ...         Yes          No        Yes
3   19  Female       Stella Maris College  ...         Yes         Yes      Maybe
4   24    Male         Guru Nanak College  ...         Yes       Maybe         No
```

df.head() is used to display the first five rows of the dataset. It helps verify that the dataset has been imported correctly and allows us to quickly preview the data without displaying the entire dataset.

```
[5 rows x 29 columns]
Age                    0
Gender                 0
College                0
Stream                 0
Currently Studying     0
Year                   0
CGPA                   0
ScreenHours            0
MainDevice             0
BeforeBed              0
Breaks                 0
SleepHours             0
Bedtime                0
SleepQuality           0
Refreshed              0
EyeStrain              0
Symptoms              28
Headache               0
EyeTired               0
HairFall6m             0
HairLoss               0
FamilyHistory          0
Stress                 0
Exercise               0
FruitsVeg              0
Water                  0
OpinionSleep           0
OpinionHair            0
OpinionEye             0
dtype: int64
```

The missing value analysis showed that the dataset contained 28 missing values, all of which were in the *Symptoms* column. No missing values were found in the other variables. To retain all 150 responses, the missing values were replaced with "None" during the data cleaning process.

```
MISSING VALUES AFTER CLEANING
Age                     0
Gender                  0
College                 0
Stream                  0
Currently Studying      0
Year                    0
CGPA                    0
ScreenHours             0
MainDevice              0
BeforeBed               0
Breaks                  0
SleepHours              0
Bedtime                 0
SleepQuality            0
Refreshed               0
EyeStrain               0
Symptoms                0
Headache                0
EyeTired                0
HairFall6m              0
HairLoss                0
FamilyHistory           0
Stress                  0
Exercise                0
FruitsVeg               0
Water                   0
OpinionSleep            0
OpinionHair             0
OpinionEye              0
dtype: int64

Data cleaned successfully!
```

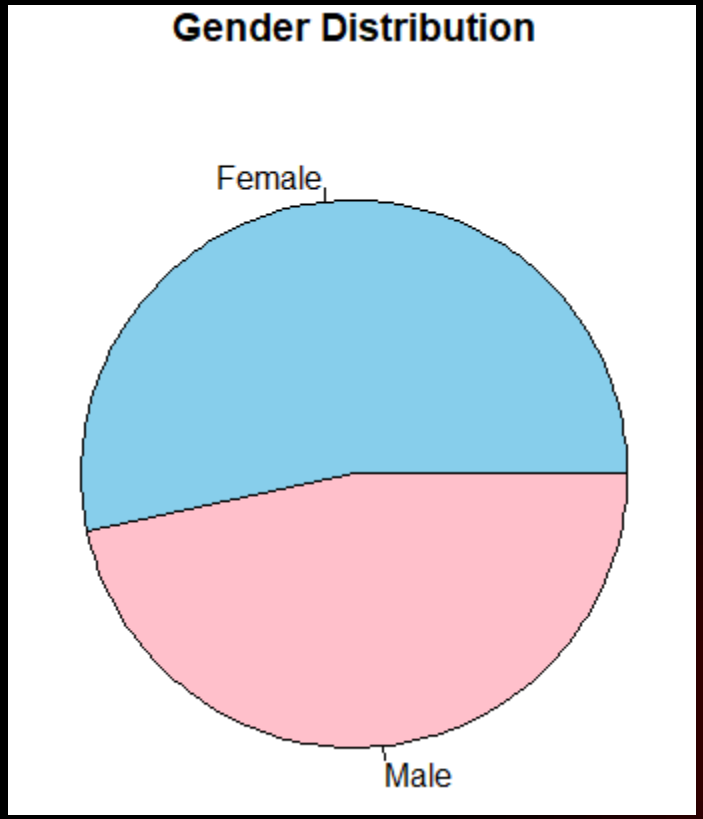
The 28 missing values in the Symptoms column were replaced with "None" using the *fillna()* function, resulting in a dataset with no missing values.

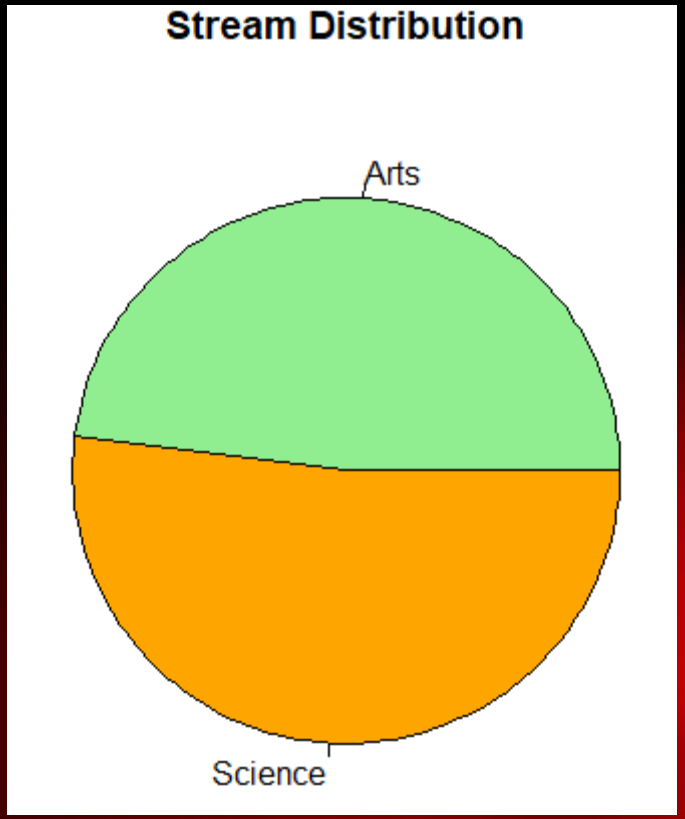
Frequency:

A frequency table is used to summarize the data by showing how many participants fall into each category. It helps us understand the distribution of the data and provides a clear overview before performing further statistical analysis

```
> # Important Frequency Tables
> freq(data$Gender)
       Frequency Percentage
Female        80      53.33
Male          70      46.67
> freq(data$Stream)
        Frequency Percentage
Arts           72         48
Science        78         52
> freq(data$ScreenHours)
    Frequency Percentage
<2         33      22.00
>8         26      17.33
2-4        27      18.00
4-6        36      24.00
6-8        28      18.67
> freq(data$SleepHours)
    Frequency Percentage
<5         25      16.67
>8         32      21.33
5-6        28      18.67
6-7        26      17.33
7-8        39      26.00
> freq(data$EyeStrain)
           Frequency Percentage
Always            30      20.00
Never             32      21.33
Often             27      18.00
Rarely            26      17.33
Sometimes         35      23.33
> freq(data$HairLoss)
                  Frequency Percentage
Mild hair loss           42      28.00
Moderate hair loss       33      22.00
No hair loss             35      23.33
Severe hair loss         40      26.67
>
```

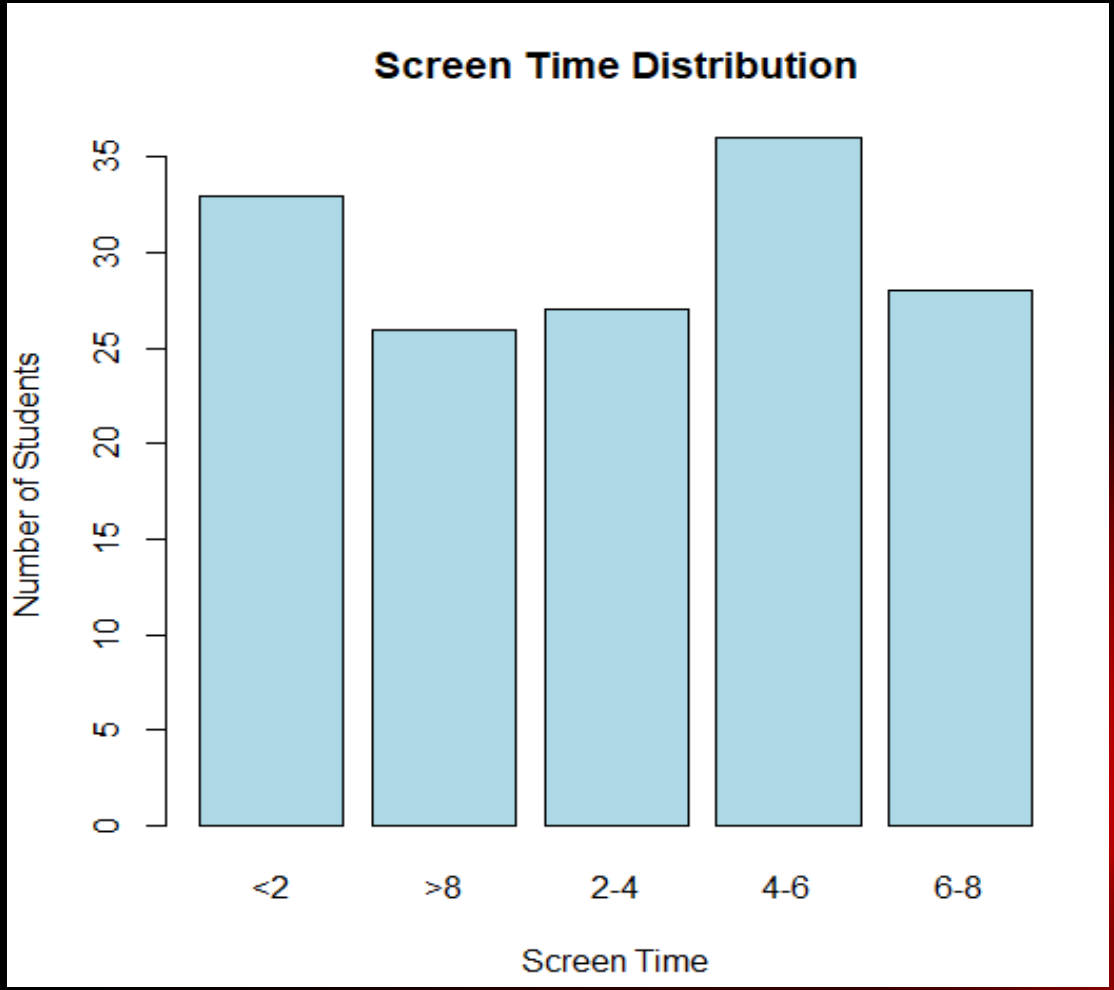
Pie Chart:


Gender Distribution
Female
Male

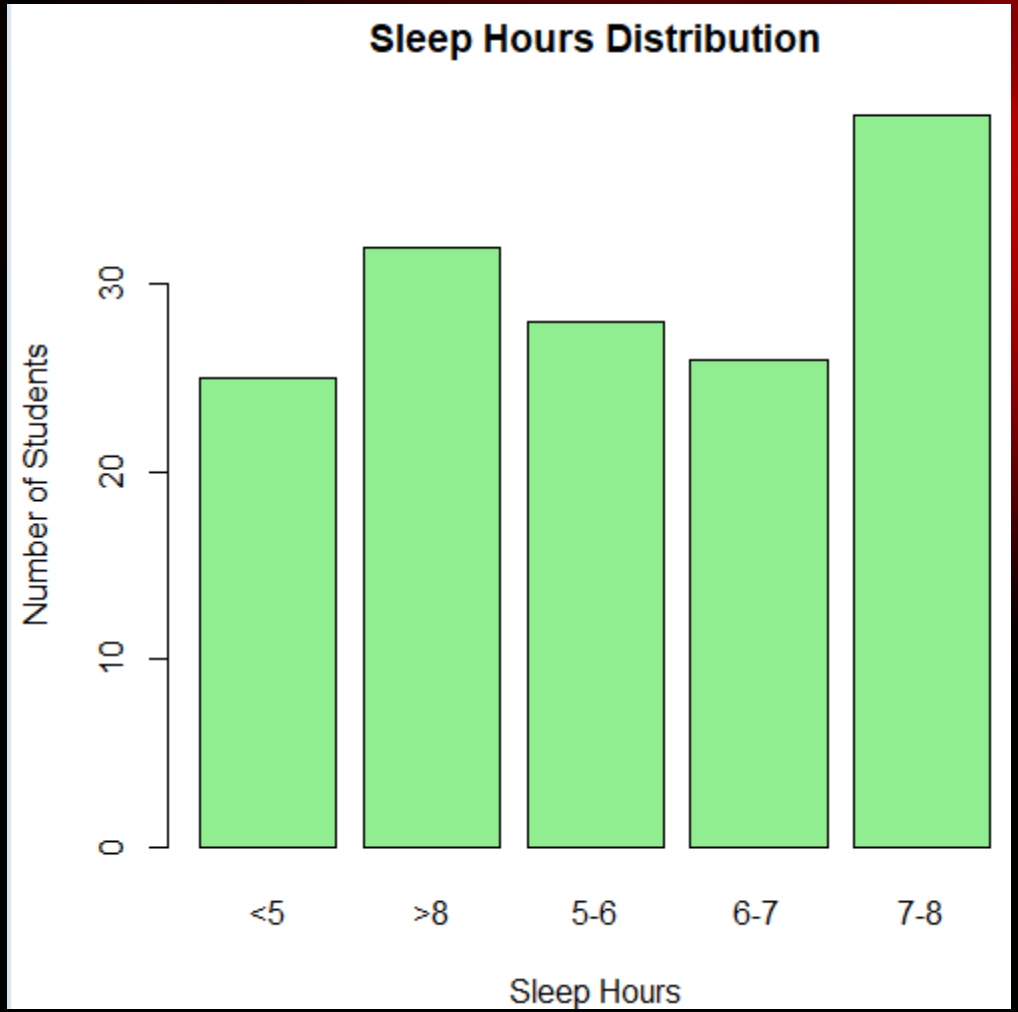

Stream Distribution
Arts
Science

Bar Diagram:

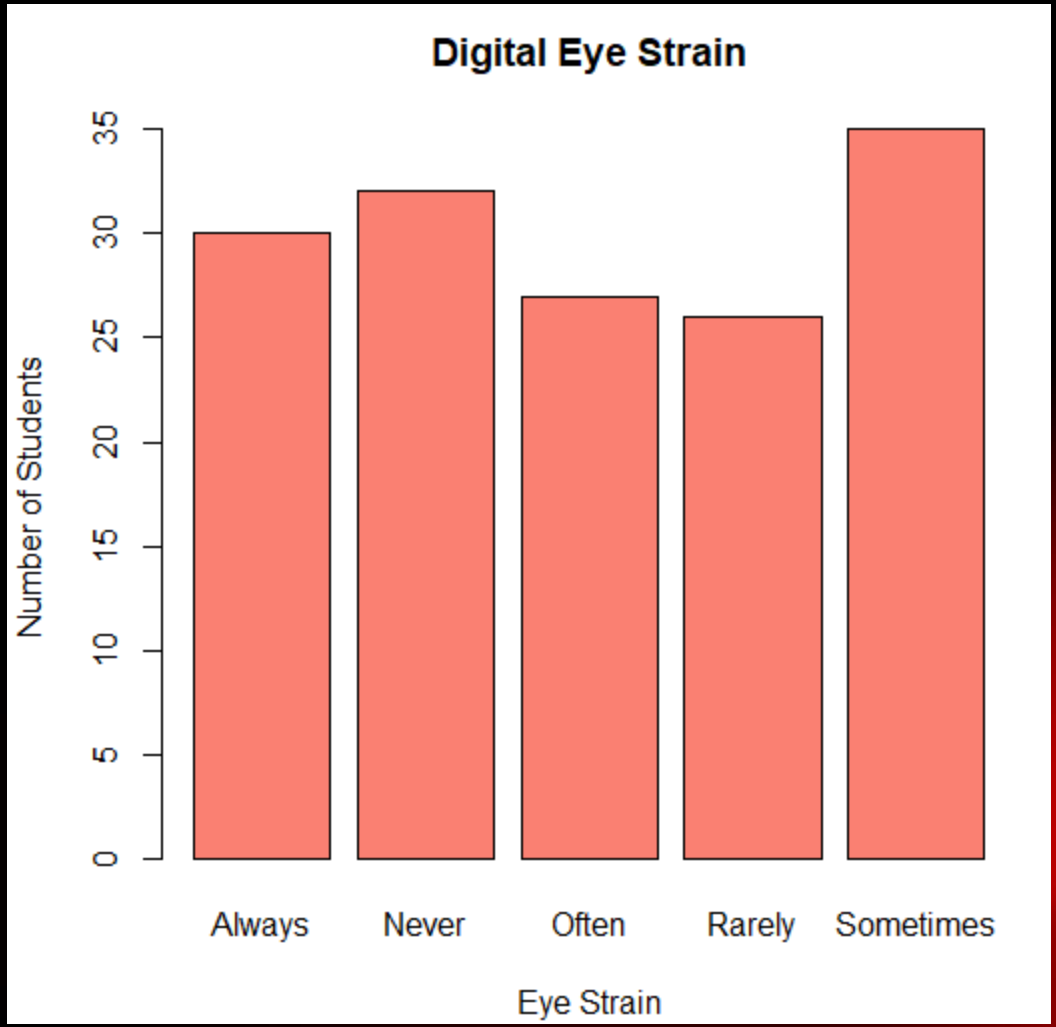
Bar Diagram between students and screen time using R



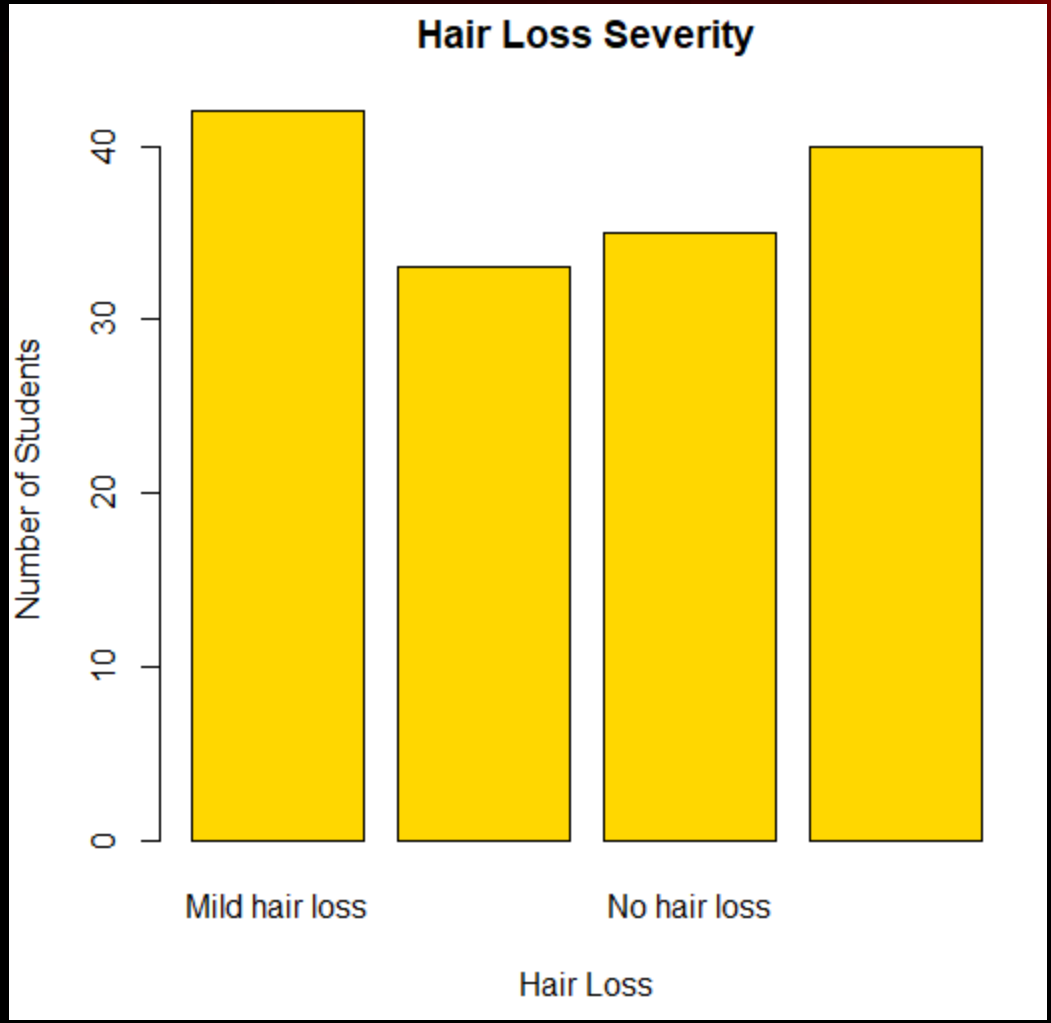
Bar Diagram between students and sleep hours using R



Bar Diagram between students and eye strain using R



Bar Diagram between students and hair loss using R



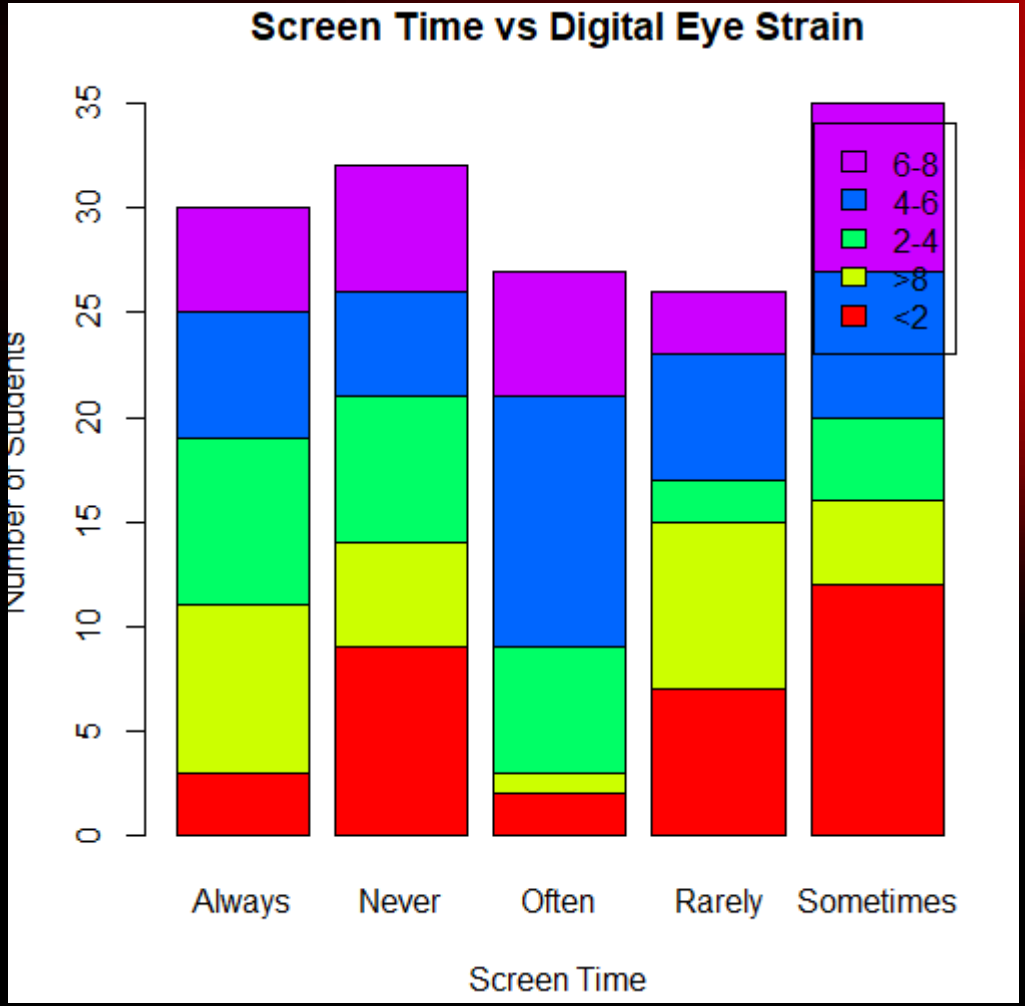
Stacked Bar Diagram:

The stacked bar chart illustrates the distribution of different sleep duration categories among students with different levels of hair loss. Each colour represents a sleep duration category. Students reporting mild and moderate hair loss include a noticeable proportion of individuals sleeping less than 5 hours or 5–6 hours, whereas students with no hair loss are more commonly found in the 7–8 hours sleep category. This indicates that shorter sleep duration may be associated with increased hair loss.



Stacked Bar Diagram:

The stacked bar chart shows the distribution of different screen time categories among students with different levels of digital eye strain (Always, Never, Often, Rarely, and Sometimes). Each coloured segment represents a screen time category. The chart indicates that students who reported 'Sometimes' and 'Always' experiencing eye strain include a considerable number of students with higher screen time (6–8 hours and more than 8 hours). This suggests a possible association between prolonged screen time and digital eye strain.



Correlation:

```
> cor(data$ScreenTimeNum, data$EyeStrainNum, method = "spearman")
[1] 0.09023894
> cor(data$ScreenTimeNum, data$HairLossNum, method = "spearman")
[1] -0.02523418
> cor(data$SleepNum, data$EyeStrainNum, method = "spearman")
[1] -0.09729912
> cor(data$SleepNum, data$HairLossNum, method = "spearman")
[1] -0.1135273
```

Spearman's correlation analysis was performed to examine the relationships among screen time, sleep hours, digital eye strain, and hair loss. The results showed very weak correlations between all pairs of variables ($|r_s| < 0.20$), indicating that there was little to no association between these variables in the present dataset.

The students in this dataset do not show a strong connection between screen time, sleep habits, eye strain, and hair loss. Even though the relationships are in the expected direction (more screen time → slightly more eye strain, more sleep → slightly less eye strain and hair loss), they are too weak to be considered strong associations

Testing using Chi-Square:

Screen Time vs Digital Eye Strain

```
> # Create contingency table
> table1 <- table(data$ScreenHours, data$EyeStrain)
>
> # Chi-square test
> chisq.test(table1)

        Pearson's Chi-squared test

data:  table1
X-squared = 27.202, df = 16, p-value = 0.03929
```

Since 0.03929 < 0.05:
Reject the null hypothesis.
There is a statistically significant association between screen time and digital eye strain.

Interpretation:
The Chi-square test showed a significant association between screen time and digital eye strain (p = 0.039). This suggests that digital eye strain varies according to students' screen time.

Sleep Hours vs Hair Loss:

```
> table2 <- table(data$SleepHours, data$HairLoss)
>
> chisq.test(table2)

        Pearson's Chi-squared test

data:  table2
X-squared = 17.589, df = 12, p-value = 0.1288
```

Since 0.1288 > 0.05:

Fail to reject the null hypothesis.

No statistically significant association

Interpretation:

The Chi-square test showed no statistically significant association between sleep hours and hair loss (p = 0.129).

Screen Time vs Hair Loss:

```
> table3 <- table(data$ScreenHours, data$HairLoss)
>
> chisq.test(table3)

        Pearson's Chi-squared test

data:  table3
X-squared = 14.907, df = 12, p-value = 0.2466
```

Since 0.2466 > 0.05:
Fail to reject the null hypothesis.
No statistically significant association

Interpretation:
The Chi-square test showed no statistically significant association between screen time and hair loss (p = 0.247).

Sleep Hours vs Digital Eye Strain:

```
> table4 <- table(data$SleepHours, data$EyeStrain)
>
> chisq.test(table4)

        Pearson's Chi-squared test

data:  table4
X-squared = 18.821, df = 16, p-value = 0.2781
```

Since 0.2781 > 0.05:
Fail to reject the null hypothesis.
No statistically significant association.

Interpretation:
The Chi-square test showed no statistically significant association between sleep hours and digital eye strain (p = 0.278).

Regression – Multiple Linear model:

Model 1: Predicting Digital Eye Strain from Screen Time and Sleep Hours:

```
> model3 <- lm(EyeStrainNum ~ ScreenTimeNum + SleepNum, data = data)
>
> summary(model3)

Call:
lm(formula = EyeStrainNum ~ ScreenTimeNum + SleepNum, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.28553 -1.16273  0.08032  1.08621  2.27797

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    3.02146    0.38052   7.940 4.78e-13 ***
ScreenTimeNum  0.08999    0.08334   1.080    0.282
SleepNum      -0.09588    0.08320  -1.152    0.251
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.419 on 147 degrees of freedom
Multiple R-squared:  0.01708,   Adjusted R-squared:  0.003703
F-statistic: 1.277 on 2 and 147 DF,  p-value: 0.282
```

```
> summary(model3)

Call:
lm(formula = EyeStrainNum ~ ScreenTimeNum + SleepNum, data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.28553 -1.16273  0.08032  1.08621  2.27797

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.02146    0.38052   7.940 4.78e-13 ***
ScreenTimeNum 0.08999    0.08334   1.080    0.282
SleepNum     -0.09588    0.08320  -1.152    0.251
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.419 on 147 degrees of freedom
Multiple R-squared:  0.01708,   Adjusted R-squared:  0.003703
F-statistic: 1.277 on 2 and 147 DF,  p-value: 0.282
```

Regression Equation:

$$\textbf{Eye Strain} = 3.021 + 0.090(\textbf{Screen Time}) - 0.096(\textbf{Sleep Hours})$$

Interpretation

Screen Time coefficient = 0.090

As screen time increases by one category, the eye strain score increases by 0.090 **units, keeping sleep hours constant.**

p = 0.282, **so it is not statistically significant.**

Sleep Hours coefficient = −0.096

As sleep hours increase by one category, the eye strain score decreases by 0.096 **units, keeping screen time constant.**

p = 0.251, **so it is not statistically significant.**

Model Fit

$R^2 = 0.017$

This means:

Only 1.7% **of the variation in digital eye strain is explained by screen time and sleep hours together.**

Conclusion

The multiple linear regression model was not statistically significant

($p = 0.282$). **Screen time and sleep hours did not significantly predict digital eye strain in this dataset.**

Model 2: Predicting Hair Loss from Screen Time and Sleep Hours:

```
> model4 <- lm(HairLossNum ~ ScreenTimeNum + SleepNum, data = data)
>
> summary(model4)

Call:
lm(formula = HairLossNum ~ ScreenTimeNum + SleepNum, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.7449 -0.6980 -0.3304  1.3020  1.6813

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.85148    0.30117   9.468   <2e-16 ***
ScreenTimeNum -0.02348    0.06596  -0.356    0.722
SleepNum      -0.08307    0.06585  -1.261    0.209
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.123 on 147 degrees of freedom
Multiple R-squared:  0.01142,   Adjusted R-squared:  -0.002031
F-statistic: 0.849 on 2 and 147 DF,  p-value: 0.4299
```

```
> summary(model4)

Call:
lm(formula = HairLossNum ~ ScreenTimeNum + SleepNum, data = data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.7449 -0.6980 -0.3304  1.3020  1.6813

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   2.85148    0.30117   9.468   <2e-16 ***
ScreenTimeNum -0.02348    0.06596  -0.356    0.722
SleepNum      -0.08307    0.06585  -1.261    0.209
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.123 on 147 degrees of freedom
Multiple R-squared:  0.01142,   Adjusted R-squared:  -0.002031
F-statistic: 0.849 on 2 and 147 DF,  p-value: 0.4299
```

Regression Equation

$$\textbf{Hair Loss} = 2.851 - 0.023(\textbf{Screen Time}) - 0.083(\textbf{Sleep Hours})$$

Interpretation

Screen Time coefficient = -0.023

Screen time has a very small negative relationship with hair loss.

p = 0.722, **so it is not statistically significant.**

Sleep Hours coefficient = -0.083

As sleep hours increase, hair loss decreases slightly.

p = 0.209, **so it is not statistically significant.**

Model Fit

- $R^2 = 0.011$

This means:

Only 1.1% **of the variation in hair loss is explained by screen time and sleep hours together**.

Conclusion

The multiple linear regression model was not statistically significant ($p = 0.430$). **Screen time and sleep hours did not significantly predict hair loss in this dataset**.

Overall Interpretation:

The study found that although screen time was significantly associated with digital eye strain based on the Chi-square test, the correlation and regression analyses indicated that the relationships among screen time, sleep quality, digital eye strain, and hair loss were generally weak in this dataset.

This suggests that these health outcomes are likely influenced by multiple factors beyond screen time and sleep habits alone.

Conclusion:

This study examined the impact of screen time and sleep quality on digital eye strain and hair loss among college students. The findings showed that screen time had a significant association with digital eye strain, indicating that students with different levels of screen time experienced different levels of eye strain. However, no significant associations were found between sleep hours and hair loss, screen time and hair loss, or sleep hours and digital eye strain.

The correlation and regression analyses revealed only weak relationships between the study variables, suggesting that screen time and sleep hours alone are not strong predictors of digital eye strain or hair loss in this dataset. These health outcomes may also be influenced by other factors such as stress, genetics, diet, physical activity, and overall lifestyle.

Overall, the study highlights the importance of maintaining healthy screen habits and adequate sleep while suggesting that further research with larger samples and additional health-related factors is needed to better understand these relationships.

THANK YOU